

18. Project Description

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Purpose of this document A short, plain-language description of AutoCare+ for anyone seeing the project for the first time — a client, investor, or new team member. No technical detail. For that, see the rest of this vault.

What is AutoCare+?

AutoCare+ is a subscription-based car care app for Zamboanga City. Instead of only visiting a repair shop when something breaks, members pay a monthly membership and get ongoing, proactive care for their vehicle — inspections, maintenance reminders, pick-up and delivery, roadside help, and a simple health score that shows exactly how their car is doing at any time.

The problem

Most car owners only think about maintenance when something has already gone wrong. Small issues go unnoticed until they turn into expensive repairs or a breakdown on the road. And when it's time to sell a car, there's usually no real proof of how well it was maintained — just the seller's word.

The solution

AutoCare+ shifts car care from reactive to proactive. Members get:

- **Scheduled inspections and reminders**, so maintenance happens on time, before small problems become big ones
- **A Vehicle Health Score** — a simple number and star rating that sums up the car's condition, with plain-language explanations for every finding
- **A "What Needs Attention" home screen**, so members always know exactly what to do next, without digging through menus
- **Pick-up and delivery**, so getting a car serviced doesn't mean losing a day
- **Roadside assistance**, tracked live in the app, for breakdowns and emergencies
- **A digital service history** tied to the vehicle, replacing paper receipts and giving buyers real proof of maintenance at resale time

Who it's for

- **Vehicle owners** in Zamboanga City who want convenient, predictable car care and peace of mind
- **Busy professionals and multi-car families** who value not having to personally drop off and pick up a vehicle
- **Fleet operators** (delivery fleets, jeepney cooperatives) who need multiple vehicles maintained reliably

How members pay

Membership plans are priced monthly, with everything included clearly listed up front. Payment can be made in cash or through common e-wallets and cards (GCash, Maya, credit/debit) — whichever a member prefers.

What makes this different

Traditional repair shops are reactive and transactional — a customer shows up once something is already wrong. AutoCare+ is a relationship: continuous care, a digital record that follows the vehicle, and a health score that turns “trust me, it’s well maintained” into something a buyer can actually verify.

Where things stand

The full requirements, technical design, and visual walkthroughs for this project are complete and documented. The application flow and technology choices are covered in detail in 16 Application Flow Diagram and 17 Technology Architecture Diagram, for anyone who wants to go one level deeper than this summary.

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16. Application Flow Diagram

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Purpose of this document This is a **visual walkthrough** of how a member moves through the AutoCare+ app, from installing it to sharing a Vehicle Health Score certificate. It is drawn from the full requirements and use cases already documented in 03 Functional Requirements and 06 Use Cases, simplified into diagrams that don't require a technical background to read. If you're presenting this to a client or stakeholder, start here.

16.1 How to read these diagrams

- **Boxes** are a screen, action, or system step.
- **Arrows** show what happens next.
- **Diamond shapes** are a decision point — the app or the member is choosing between paths.
- **Colour** groups steps that belong to the same stage or lane; it has no other meaning.
- **Dashed arrows** show an optional or secondary path (for example, looping back to book a repair after checking the dashboard).

16.2 Diagram 1 — End-to-End Journey Overview

This is the big picture: five stages every member passes through, from download to after-service.



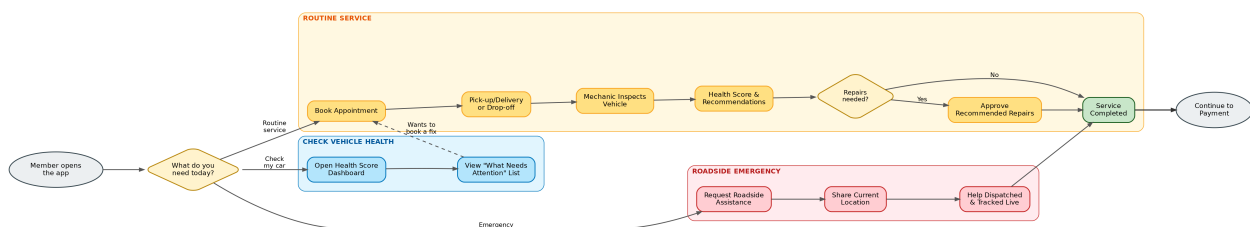
What's happening at each stage

Stage	What the member experiences	Source requirements
1 • Getting Started	The member downloads the app, verifies their phone number with a one-time code (OTP), agrees to the privacy policy, and registers their vehicle's basic details.	03 Functional Requirements → 3.1 M1 — Identity & Accounts
2 • Choose a Plan	The member compares the Basic and Premium membership plans, picks how they'll pay (cash or e-wallet), and their subscription goes live.	03 Functional Requirements → 3.2 M2 — Subscription & Billing

Stage	What the member experiences	Source requirements
3 • Using the App	This is where members spend most of their time — booking services, checking their car's health, or requesting emergency help. Diagram 2 below breaks this stage down in detail.	03 Functional Requirements → 3.4 M4 — Scheduling & Capacity
4 • Payment & Records	Once a service is done, the member pays (in cash or through GCash/Maya/card), gets a digital receipt, and the visit is automatically added to their vehicle's permanent service history.	03 Functional Requirements → 3.8 M8 — Payments
5 • After Service	The member gets their vehicle back, and can optionally generate a shareable Health Score certificate — useful proof of maintenance if they ever sell the car.	03 Functional Requirements → 3.5 M5 — Inspection & Vehicle Health Score

16.3 Diagram 2 — Core In-App Service Flow

This zooms into **Stage 3** above — the three things a member can do inside the app, and how each one unfolds.



The three paths, explained

Routine Service (yellow) The most common path. The member books a service, chooses whether AutoCare+ picks up their car or they drop it off, a mechanic inspects it, and the app generates a **Vehicle Health Score** with any recommended repairs. If repairs are needed, the member approves them individually before any work happens — nothing is done to their car without their sign-off. Full detail: UC-003 and UC-006 in 06 Use Cases → 6.1 Actor-goal list, and 06 Use Cases → UC-005 — Perform a vehicle inspection.

Roadside Emergency (red) For breakdowns — a flat tyre, dead battery, or being stranded on the road. The member shares their location, and help is dispatched and tracked live in the app, the same way a ride-hailing app tracks a driver. Full detail: 06 Use Cases → UC-007 — Request roadside assistance.

Check Vehicle Health (blue) The member opens their dashboard at any time — not tied to a booking — to see their car’s current health score and a **“What Needs Attention”** list: a plain summary of anything that needs looking at. From there, they can jump straight into booking a fix. Full detail: 03 Functional Requirements → 3.11 M11 — Attention Dashboard & Component Visualization.

16.4 Handoff Notes — Read This Before Sharing With Your Client

Purpose of this section This section exists so that when this document is shared with a non-technical client, they can follow it without needing you in the room to translate. It defines every term a client is likely to stumble on, and flags what’s built now versus later.

Plain-language glossary for this diagram

Term in the diagram	What it means, in plain terms
OTP	“One-Time Password” — a temporary code sent by text message to confirm it’s really the member signing up. Same thing used by banking apps.
Vehicle Health Score (VHS)	A single number (and star rating) summarizing how healthy the car is, generated automatically after every inspection — like a report card for the vehicle.
“What Needs Attention” list	A simple, always-visible list on the home screen telling the member exactly what to fix and why, instead of them having to dig through menus.
Entitlement / plan quota	What’s included in a membership plan each month (for example, “1 inspection per month”). Doesn’t appear directly in this diagram, but governs what “Book Appointment” allows.
E-wallet	A digital payment app like GCash or Maya — the Philippine equivalent of a mobile wallet.
Digital service history	A permanent online record of everything ever done to the vehicle, replacing paper receipts.
Certificate	A shareable web link / PDF showing the car’s health score and service history — useful when selling the car, since it proves it was properly maintained.

What’s live now vs. planned later

Item	Status
Everything shown in both diagrams above	Included in the first release (v1.0)

Item	Status
A visual, interactive diagram of the car itself (tap a part of the car to see its condition)	Planned for a later release (v1.1) — not shown here since it isn't part of the first version. See 13 Constraints and Risks → Roadmap — visual damage diagram (formerly “3D Repair Visualization”) for why.

Questions a client will likely ask, answered in advance

“Does the client have to approve every single repair?” Yes — see the “Repairs needed?” decision point in Diagram 2. No work happens on a member’s car without them explicitly approving it in the app first. This is a deliberate trust-building feature, not a limitation.

“What happens if the member doesn’t have a smartphone or data connection?” The diagrams assume a smartphone with mobile data, which is the target user for a mobile-first app like this. Members without a compatible device would need to visit or call the physical shop directly — that fallback isn’t part of the app itself.

“Can a member pay in cash, or is it card-only?” Cash is fully supported, shown as “Pay by cash or e-wallet” in Stage 4. This was a deliberate choice — see 13 Constraints and Risks → COD versus recurring subscription for why cash matters for this market.

“Is this the final design, or will screens look different?” These are **flow diagrams**, not screen designs — they show the sequence of steps and decisions, not what the app will look like. Actual screen layouts are a separate design phase; see 12 Screen Inventory for the full list of planned screens.

Related

- 06 Use Cases — the detailed, technical version of the flows shown here
- 03 Functional Requirements — every requirement these diagrams are built from
- 17 Technology Architecture Diagram — what’s running behind the scenes to make this flow work
- 12 Screen Inventory — the actual screens that implement each step

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17. Technology Architecture Diagram

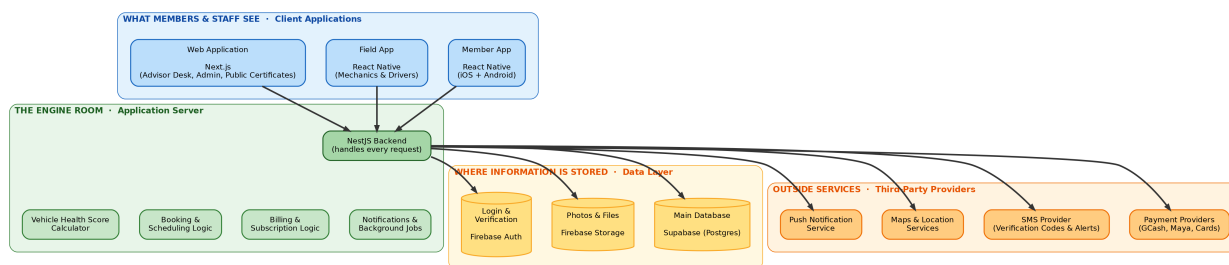
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Purpose of this document This shows **what AutoCare+ is built with**, in four plain layers, and explains why each piece was chosen. It's the visual counterpart to 07 System Architecture, which covers the same ground in full technical detail. Use this version when presenting the technology choices to a client or non-technical stakeholder.

17.1 How to read this diagram

Think of it like a building: what people see (the apps), the engine room that does the work (the server), where records are filed (storage), and outside contractors the business relies on (payment providers, SMS, maps). Every arrow means “talks to” — information flows from client apps down through the server to storage and outside services, and back.

17.2 The diagram



17.3 Technology at a glance

Each row is a tool or service, what it actually does in plain terms, and which features it powers.

Layer	Technology	Plain-English purpose	Powers these features
Client	React Native (Member App)	One codebase that produces both the iOS and Android app, instead of building two separate apps	Registration, booking, health score viewing, payments, roadside requests
Client	React Native (Field App)	The same framework, built into a separate app for mechanics and drivers, usable without an internet connection	Vehicle inspections, pick-up/delivery trip tracking, cash collection

Layer	Technology	Plain-English purpose	Powers these features
Client	Next.js (Web Application)	A website framework that powers three separate things from one codebase: the service advisor's desk screen, the admin dashboard, and the public health-certificate pages anyone can view via a shared link	Scheduling board, reporting, plan/pricing management, shareable Health Score certificates
Server	NestJS	The central program that every app (member, field, web) talks to. It enforces every business rule — who can do what, what a plan includes, how billing works — in one place, so the rules are consistent everywhere	Every feature in the app, without exception
Server	Vehicle Health Score Calculator	A dedicated piece of logic, inside the backend, that turns a mechanic's inspection into the 0-100 score and star rating members see	Health scores, "What Needs Attention" dashboard, shareable certificates
Data	Supabase (Postgres)	The main database — where every vehicle, booking, subscription, and inspection record actually lives	All records and history in the app
Data	Firebase Storage	Where photos (inspection photos, vehicle photos, documents) are kept	Inspection photo capture, document uploads
Data	Firebase Auth	Handles login and phone-number verification, including the one-time SMS codes	Sign-up, login, OTP verification

Layer	Technology	Plain-English purpose	Powers these features
Outside	Payment Providers (GCash, Maya, card processors)	Lets members pay through the e-wallets and cards Filipino consumers already use, without AutoCare+ ever handling raw card numbers	Subscription billing, work-order payments
Outside	SMS Provider	Sends the verification codes and critical alerts (e.g. “your roadside help is on the way”) as text messages, as a backup to app notifications	OTP delivery, critical alerts when the app can’t reach the phone
Outside	Maps & Location Services	Turns a GPS pin into a readable address and shows a live map of dispatched help or a delivery driver	Pick-up requests, roadside assistance tracking
Outside	Push Notification Service	Sends the everyday reminders and updates — “your appointment is tomorrow,” “your car is ready”	Appointment reminders, status updates, service-due alerts

17.4 Handoff Notes — Read This Before Sharing With Your Client

Purpose of this section Technology choices are one of the first things a client questions, especially around cost and vendor risk. This section pre-answers the likely questions in plain language, without requiring you to explain the technical architecture live.

Plain-language glossary for this diagram

Term	What it means, in plain terms
Backend / server	The “engine room” — the part of the app nobody sees directly, but which does all the actual work: checking who’s allowed to do what, calculating scores, processing payments.

Term	What it means, in plain terms
Database	Where all the information is permanently stored — every member, vehicle, booking, and inspection record. Think of it as a very large, extremely organized filing cabinet.
API	The “language” the apps use to ask the server for information or tell it to do something (e.g., “book this appointment”). Not shown as a separate box here — it’s built into the NestJS backend.
Cloud hosting	The server and database run on rented computing infrastructure over the internet, rather than on a physical computer in the shop. This is standard for any modern app and is what makes the app accessible from anywhere.
Free tier	Several of the outside tools (the database and login service specifically) offer a free usage allowance before any cost kicks in — meaningful for keeping early costs low, with a plan to upgrade as the business grows. See 13 Constraints and Risks → 13.1 Locked constraints for the specific quotas.
Third-party provider	A company AutoCare+ pays or partners with instead of building that capability itself — for example, using an existing payment company instead of building payment processing from scratch, which would be slow, expensive, and risky to get right.
Vendor lock-in	The risk of depending so heavily on one outside provider that switching later becomes difficult or expensive. Addressed directly in the architecture — see the note below.

Questions a client will likely ask, answered in advance

“Why do we need so many different outside tools instead of building everything in-house?” Each of these (payments, SMS, maps, push notifications) is a solved problem elsewhere, and building any one of them from scratch would take months and carry real risk (especially payments, which involves handling money and complying with banking regulations). Using established providers for these pieces means faster delivery and fewer places for something to go wrong, while all of AutoCare+’s actual business logic — pricing, scheduling, health scoring — stays custom-built and owned by the business.

“What if one of these companies raises their prices or shuts down?” This was accounted for directly in the design. The backend is built so that the database and login service can be swapped for a different provider without rewriting the apps — see 07 System Architecture → 7.3 Reconciling Firebase Auth with Supabase RLS for the technical detail. The system isn’t permanently tied to any one vendor.

“Does the business own the app, or does it belong to whoever builds it?” The architecture and all custom code (the NestJS server, the mobile and web apps,

the scoring algorithm) belong entirely to the business. The outside services are pay-as-you-go tools the app *uses*, the same way a shop uses a card payment terminal without owning the card network.

“Is customer data safe?” Yes — this is covered in full in 10 Security and Privacy, including compliance with the Philippine Data Privacy Act. In short: passwords and sensitive data are encrypted, card numbers never touch AutoCare+’s own servers, and every sensitive action is logged.

“Why two mobile apps instead of one?” Members and staff need very different things. Members need a polished, simple app. Mechanics and drivers need something that works reliably with a spotty signal in the field and survives being used one-handed while holding a tool. Separating them keeps each app focused and easier to use for its actual audience — see 02 Overall Description → 2.2 User classes and characteristics.

Related

- 07 System Architecture — the full technical version of this diagram, with implementation detail
- 08 Data Model — what’s actually stored in the database shown here
- 10 Security and Privacy — how member and vehicle data is protected
- 13 Constraints and Risks — costs, quotas, and vendor risk in detail

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